

Assignment 01

1. Suppose that DS = 0250H, BX = 0370H, and DI = 400H . Determine the memory address accessed by each of the following instructions, assuming real mode operation:

- (a) MOV AL, [WXYZ H]
- (b) MOV [DI], AL
- (c) MOV DL, [BX+ DI]

Where WXYZ are the last four digits of your ID.

2. In the real mode, show the starting and ending addresses of each segment located by the following segment register values:

- (a) 1000H
- (b) 1234H

3. Convert the following:

- (a) MOV BX, SS: [BP+DI] from assembly language to machine language.
- (b) 8B07H from machine language to assembly language.

4. Suppose, Core2 microprocessor operates in protected mode into Windows OS. To perform read or write operation every microprocessor needs to calculate the physical address of RAM/ROM & in protected mode the segment register and descriptor table are used for this purpose. Now consider the following scenario:

- i. Define the address bit of the descriptor so that microprocessors use up to 4GB of the RAM.
- ii. By default, you can use 64KB or maximum 16MB for any segment (DS/CS/SS/ES) portion of the RAM.
- iii. You must define the descriptor privilege level of access right portion so that segment access will be granted.
- iv. Set a bit of access right portion that ensures CS (code segment) has valid address and limit.
- v. Also ensure the descriptor describes the code segment portion.
- vi. Here, confirm that code segment working through the privilege level that was previously defined.
- vii. Define a bit in the access right portion that ensures if necessary code segments may be read.
- viii. You must define a bit in the descriptor so that segment extension feature will be accepted.

ix. There will be a bit in the descriptor, which will ensure that the microprocessor uses 32 bit instructions mode because of working in Windows OS.

x. For the confirmation of Core2 working with 32-bit compatibility mode, there also define a bit in the descriptor.

xi. In order to ensure code segment (CS) is available there should be defined a bit in the descriptor.

Your task is to define 64 bits of the descriptor format used for the Core2 microprocessor. Also analyze the details of each bit so that the above operations will perform correctly.